

The affine cipher or times cipher needs the times part of the encryption to be co-prime with 26 for the decryption to be consistent.

When the times is not co-prime with 26, there is 1/2 of the possibilities for the ct letters, and therefore a different distribution of pattern types among the ct letters.

Consider a n-nested keyed vigenere cipher encrypt with alphabet key kryptos
passphrase ABCDEFGHIJKLMNOPQRSTUVWXYZ

nested# ct

```
01 ABCDEFGHIJKLMNOPQRSTUVWXYZ=pt
02 HJMQVXKYTSKBDFDSHYFBJMQVTX
03 UXRTADGJNVKZY0ICBPMFEHLQSW
04 YSDHMKVBTBFKJQXQFYTXJSDHMBV
05 CHNXPBGMWYKAFLZIVOTQRSEJDU
06 JVYBHQKSFMKXTDTMJSDXVYBHFQ
07 WTEMZSGQRBKIVPCUDAJYLYXOFHN
08 TFQYDMKBJXKSHVHXTBVSFQYDJM
09 EQPFUTGVOHKWSINRXCDAJZBML
10 MYFVSJKDQTKHXBXTMDBHYFVSQJ
11 ZDUSLYGXCQKOJRPAFEHMBNTVI
12 SMTJYHKFXDKVBQBDSFQVMTJYXH
13 GKGKGKGKGKGKGKGKGKGKGKGKGKG
14 QBVDXFKHYJKTMSMJQHSTBVDXYF
15 RJOQCXGYLSKUDZWNHIFBPMVTE
16 BXHTQDKJSVKFYMYVBJMFXHTQSD
17 ISWHOVTUFGPKQEAZYLXJNDRMBC
18 VHSXJBKMDYKQFTFYMTQHSXJDB
19 PVIBRQGSZMKETWLOJNDXCYUHFA
20 DTXMFSKQHBKYVJVBDQJYTXMFHS
21 LFAYWMGBPKNHCRETUVSZQIDJO
22 XQJFBTKVMHKDSYSHXVYDQJFBMT
23 OYZVNJGDATKRXUELMBHIFCSQP
24 FDBSTYKXVQKMJHJQFXHMDBSTVY
25 NMLJIHGFEDKCAUWSZQVOTPYXR
26 KKKKKKKKKKKKKKKKKKKKKKKKKKK 26=0
27 ABCDEFGHIJKLMNOPQRSTUVWXYZ 27=1
```

This is modified affine cipher encrypt with alphabet key kryptos and $c = \text{nested} * p = (\text{nested} - 1) * p + p$, where nested varies from 1 to 26, (for usual affine, $c = a * p + b$, a and b are fixed).
When nested=2, $c = 2 * p = 1 * p + p = p + p$ so clearly there will be 2 matches for this case.

For this case, notice that
 $2 * \text{LMNABGCDJIEFH} = \text{BDFHJMKQSTVXY}$ and
 $2 * \text{TOSQUKVWPYXZR} = \text{BDFHJMKQSTVXY}$

Therefore, there is no solution p for $c = 2 * p$ when c is any of the letters CEGILRNUAOWZP, so only YVTTMFK but not NPZP in NYPVTTMZFPK can be encrypted this way.

For example,
 $2 * I = T$, but also $2 * Y = T$; and
 $2 * C = M$, but also $2 * V = M$.
 $p_{1/2}$

In other words, for decryption, only 1/2 of the c characters can be decrypted at all, and for each one there is 2 choices for the p characters. These points should be factored into the probability calculation for 2 match odds.

Now, if nested is not fixed at 2 and instead is allowed to vary, we very quickly notice a pattern that at first glance seems to be anything but coincidence.

$N = E + B = \text{none} * B + B$, however,

$M = D + B = 11 * B + B = 12 * B$
 $Y = L + E = 11 * E + E = 12 * E$
 $P = Y + R = 02 * R + R = 03 * R$
 $V = O + L = 11 * L + L = 12 * L$
 $T = I + I = 01 * I + I = 02 * I$
 $T = E + N = 17 * N + N = 18 * N$
 $M = C + C = 01 * C + C = 02 * C$
 $Z = B + L = 02 * L + L = 03 * L$
 $F = A + 0 = 17 * 0 + 0 = 18 * 0$
 $P = Q + C = 08 * C + C = 09 * C$
 $K = K + K = \text{any} * K + K = \text{any} *$

The pattern is:

AABACDCBDE?
**_ *-----
--*-----*--
----*-*-----
-----*--*--
-----*--
-----*--
-----*--

where doublets (**) are very close together where ? is any and

$A = 12 = 2^2 * 3^1 = \{2, 3\}^{\{2, 1\}}$
 $B = 03 = 2^0 * 3^1 = \{2, 3\}^{\{0, 1\}}$
 $C = 02 = 2^1 * 3^0 = \{2, 3\}^{\{1, 0\}}$
 $D = 18 = 2^1 * 3^2 = \{2, 3\}^{\{1, 2\}}$
 $E = 09 = 2^0 * 3^2 = \{2, 3\}^{\{0, 2\}}$
 $F = 04 = 2^2 * 3^0 = \{2, 3\}^{\{2, 0\}}$

Here $? = F$ to make it more of a pattern, the powers of 2 and 3 are circular shifts of each other! That is, both are circular shifts of 011022.

Any way I have tried to continue the pattern to the next 11 characters WGDKZXTJCDI I have not been able to get any words from it.

Moreover, there is still the problem of decryption giving 2 possible pt characters for 1/2 of the ct characters for all nests of those characters even when using the trick we used with the N (no EN under B, but there is a DM under B).

So in the end we have 2^{11} choices to check for the next 11 characters for each of the ways we try to continue the pattern.

However, the artist of Kryptos may have found a way to modify the affine cipher or times cipher so that decryption is consistent.